BCA 4-Year On-Job Training (OJT) Official Plan & Logbook

SPU BCA SEMESTER 7 • COMPULSORY 6 CREDITS SPECIALIZATION DOSSIER (YEAR 2026)

SPU Compulsory: Sem 7 • 6 Credits

Calendar: 12 Oct 2026 – **24 Nov 2026 (SPU Deadline)**

SPU BCA OJT Compliance: As per SPU circular, Sem 7 OJT mandates **30 Days / 180 Hours (6 Credits)** before **24 Nov 2026 deadline**. Supervised by Company Mentor & Faculty Coordinator. Mandates Attendance, Daily Logbook, Capstone Project, 10-Section Report Dossier, Presentation Deck & Final Viva Voce Defense.

DOMAIN SPECIALIZATION	DASHBOARD TOOL STACK	CERTIFIED HOURS TARGET	SPU FINAL DEADLINE
Data Analysis (Python & SQL)	**Streamlit Cloud + Plotly Web App**	**180.0 Hours (6.0 hrs / day strictly)**	**24 November 2026 (Hard Deadline)**

44 Days	32 Days	6.0 Hours	180.0 Hours
TOTAL DURATION WINDOW	ACTIVE WORKING DAYS	DAILY OJT COMMITMENT	CERTIFIED HOURS TARGET

Day 1–22 Model: Theory (2.0h) • Hands-on Lab (2.5h) • Logbook/Practice (1.5h) •**Last 10 Days Model (Day 23–32):** **Dashboard Making Skill (2.0h)** • **Project Development (4.0h)**

DATE & DAY (2026)	TRAINING STATUS / TOPIC	DAILY WORK REPORT LOG (THEORY + HANDS-ON LAB / DASHBOARD SKILL + PROJECT)	HOURS LOGGED
12 Oct 2026 Mon (Cal Day 1)	WORKING DAY 1 **Python Data Stack Setup & NumPy Foundations**	**THEORY & CONCEPT (2.0H):** Python data science ecosystem ka overview samjha; VS Code aur Jupyter Notebook kernel configuration seekhi. Python native lists (pointer-based) aur NumPy contiguous memory buffer (C-level execution speed) ke beech performance difference study kiya. Array dimensionality (1D, 2D, 3D), ndarray attributes (shape, size, ndim, dtype) aur numeric memory allocations (int32 vs float64) ka theoretical concept clear kiya. **HANDS-ON LAB & CODE (4.0H):** Jupyter lab me numpy initialize karke `np.array()`, `np.zeros((3,4))`, `np.ones((2,5))`, `np.arange(10,50,2)`, aur `np.linspace(0,1,100)` execute kiye. Created arrays ke `.shape`, `.dtype`, `.itemsize`, aur `.nbytes` print karke memory consumption audit verify kiya. Day 1 ka environment validation log complete kiya.	**6 hrs** Certified: **6 hrs**
13 Oct 2026 Tue (Cal Day 2)	WORKING DAY 2 **NumPy Indexing, Slicing & Boolean Masking**	**THEORY & CONCEPT (2.0H):** Multi-dimensional arrays me data extraction aur slicing mechanics ko deeply study kiya. Memory View vs Deep Copy ka concept samjha ki basic slicing view return karti hai jisse original memory link rehti hai. Conditional filtering ke liye Boolean Masking ka mathematical logic samjha, jisme bitwise operators (&, |, ~) ke through complex logic execute hoti hai. **HANDS-ON LAB & CODE (4.0H):** 5x5 numeric matrix create karke row-column sub-setting ki (`arr[1:4, 2:5]`, step strides `arr[::2, ::-1]`). Condition-based filtering execute ki jaise `arr[arr > 50]` aur multiple boolean criteria evaluate kiye. `np.where()` function use karke conditional value replacement implement kiya. 10 practical matrix extraction problems solve kiye.	**6 hrs** Certified: **12 hrs**
14 Oct 2026 Wed (Cal Day 3)	WORKING DAY 3 **Array Manipulation, Reshaping & Stacking**	**THEORY & CONCEPT (2.0H):** Array structural reorganization aur dimension manipulation algorithms padhe. `.reshape()` execute karne ke rules (size consistency) aur Row-Major (C-style) vs Column-Major (Fortran-style) memory layout samjha. Multidimensional array ko 1D me transform karne ke liye `.flatten()` (deep copy) aur `.ravel()` (memory reference) ke beech architectural difference study kiya. **HANDS-ON LAB & CODE (4.0H):** 24 elements ke 1D array ko (2,3,4), (4,6), aur (8,3) shape matrices me transform kiya. Multi-feature dataset simulation ke liye `np.vstack()`, `np.hstack()`, `np.dstack()`, aur `np.concatenate()` execute karke vertical aur horizontal array combining ki. `np.split()` aur `np.hsplit()` se arrays ko train-test splits jaisi chunks me divide kiya.	**6 hrs** Certified: **18 hrs**
15 Oct 2026 Thu (Cal Day 4)	WORKING DAY 4 **Vectorization, Broadcasting & Mathematical Operations**	**THEORY & CONCEPT (2.0H):** Vectorization aur Broadcasting Rules ka mathematical framework samjha. Traditional Python loops ke comparison me vectorized operations C-loops par fast execute hote hain. Broadcasting ke dimensions rules (trailing axes equality ya dimension 1) samjhe. Universal functions (ufuncs) aur axis-based summary statistics (`axis=0` columns, `axis=1` rows) ka concept clear kiya. **HANDS-ON LAB & CODE (4.0H):** Incompatible shapes jaise (4,3) matrix aur (1,3) vector par arithmetic broadcasting test kiya. Statistical functions execute kiye: `np.mean()`, `np.median()`, `np.std()`, `np.var()`, aur `np.percentile()` along specific axes. Linear algebra me dot product (`np.dot`), matrix multiplication (@), aur transpose (`.T`) verify kiya.	**6 hrs** Certified: **24 hrs**
16 Oct 2026 Fri (Cal Day 5)	WORKING DAY 5 **Random Sampling & Introduction to Pandas**	**THEORY & CONCEPT (2.0H):** Probabilistic random sampling aur reproducibility ke liye `np.random.seed()` ka role samjha. Core analytics library Pandas ka introduction start hua. Tabular business data store karne ke liye do primary data structures: 1D labeled `Series` aur 2D tabular grid `DataFrame` ki architecture study ki. Explicit Index vs Implicit Range Index mechanics samjhe. **HANDS-ON LAB & CODE (4.0H):** `np.random.normal()`, `np.random.randint()`, aur `np.random.choice()` use karke 1,000 synthetic transaction records generate kiye aur binary `.npy/.npz` me save/load kiya. In arrays ko Python dictionary me pack karke pehla Pandas DataFrame construct kiya. Index attributes, column headers, aur `.dtypes` inspect kiye.	**6 hrs** Certified: **30 hrs**

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17 Oct 2026 Sat (Cal Day 6)	WORKING DAY 6 **Pandas Data Ingestion & Structural Inspection**	**THEORY & CONCEPT (2.0H):** External data sources se tabular data ingest karne aur data integrity verify karne ke methods explore kiye. Alag-alag file formats (CSV, Excel `.xlsx`, JSON) ke ingestion parameters jaise `encoding='utf-8'`, `delimiter`, `na_values`, aur `parse_dates` study kiye. Large datasets ke memory footprint ko evaluate karne ki importance samjhi. **HANDS-ON LAB & CODE (4.0H):** 50,000+ records ka Telecom Transaction CSV dataset Pandas me ingest kiya (`pd.read_csv()`). Structural inspection functions run kiye: `.head(10)`, `.tail(10)`, `.shape`, `.columns`, aur `.info()` jisse null counts aur dtypes verify huye. `.describe()` run karke mean, std dev, min, 25%, 50%, 75%, max values ka summary extract kiya.	**6 hrs** Certified: **36 hrs**
18 Oct 2026 Sun (Cal Day 7)	SUNDAY HOLIDAY	***Weekly Academic Off:*** *Self-study, revision of previous week concepts & SPU logbook maintenance.*	**0 hrs** Certified: 36 hrs
19 Oct 2026 Mon (Cal Day 8)	WORKING DAY 7 **Data Selection & Querying with .loc and .iloc**	**THEORY & CONCEPT (2.0H):** Pandas me precise row/column indexing aur sub-setting techniques study ki. Do core indexers ka fundamental difference samjha: **`.loc[]`** jo purely label-based hota hai (index name aur column header ke basis par filter karta hai) aur **`.iloc[]`** jo integer position-based (0-indexed position) hota hai. `df.query()` syntax aur boolean mask efficiency compare ki. **HANDS-ON LAB & CODE (4.0H):** `.loc` ke through specific date ranges aur region columns extract kiye, jabki `.iloc` se top 100 rows aur positional slices extract kiye. Multi-condition queries run ki jaise: amount > 500 aur status 'SUCCESS' (`df[(df['amount'] > 500) & (df['status'] == 'SUCCESS')]`). `.isin()` aur `.str.contains()` filtering use karke 12 practical business scenarios solve kiye.	**6 hrs** Certified: **42 hrs**
20 Oct 2026 Tue (Cal Day 9)	DUSSEHRA (VIJAYADASHAMI)	***Academic / Public Off:*** *No company training sessions scheduled (Dussehra (Vijayadashami)).*	**0 hrs** Certified: 42 hrs
21 Oct 2026 Wed (Cal Day 10)	WORKING DAY 8 **Data Cleaning: Missing Values & Duplicates**	**THEORY & CONCEPT (2.0H):** Data quality audit aur cleaning pipeline methodology study ki. Missing data (NaN, None, NaT) ke causes samjhe: MCAR (Missing Completely at Random) vs MNAR (Missing Not at Random). Drop strategy (`dropna()`) kab use karni hai aur Statistical Imputation (`fillna()` with Mean for normal distribution, Median for skewed, Mode for categorical) ka decision framework establish kiya. **HANDS-ON LAB & CODE (4.0H):** Dataset par `.isna().sum()` aur `.isnull().mean() * 100` run karke missing % report banayi. Critical columns me jahan user ID missing thi drop kiya; delivery/recharge fee me median value impute ki. Duplicate records identify karne ke liye `df.duplicated()` run kiya aur `drop_duplicates(subset=['transaction_id'], keep='first')` execute karke 100% data integrity achieve ki.	**6 hrs** Certified: **48 hrs**
22 Oct 2026 Thu (Cal Day 11)	WORKING DAY 9 **Type Casting & Vectorized Text Normalization**	**THEORY & CONCEPT (2.0H):** Memory optimization aur string cleaning techniques study ki. Uncleaned datasets me numeric prices aur dates aksar object (string) format me aate hain jisse computation error hota hai. `astype()` aur `pd.to_numeric(errors='coerce')` ke underlying behavior ko samjha. Vectorized `.str` accessor methods aur regex pattern substitution ke principles padhe. **HANDS-ON LAB & CODE (4.0H):** Price column se currency symbols (₹, commas) strip karke `float64` me cast kiya. Categorical columns (Operator, Plan Type, Circle) par `.str.strip()`, `.str.lower()`, aur `.str.title()` standardise kiye. Categorical text columns ko `category` dtype me convert karke overall DataFrame memory consumption 40% reduce kiya.	**6 hrs** Certified: **54 hrs**
23 Oct 2026 Fri (Cal Day 12)	WORKING DAY 10 **Pandas GroupBy & Multi-Level Aggregations**	**THEORY & CONCEPT (2.0H):** Data analytics ka core pattern—**Split-Apply-Combine Paradigm**—deeply study kiya. Dataset ko categorical keys par split karke summary statistics (Sum, Mean, Count, Min, Max) compute karne ka workflow samjha. `.agg()` method ke through multiple columns par simultaneous custom aggregations aur lambda functions apply karne ke rules clear kiye. **HANDS-ON LAB & CODE (4.0H):** Operator-wise total sales: `df.groupby('operator')['amount'].sum()` compute kiya. Multi-level grouping execute ki: `df.groupby(['operator', 'circle']).agg({'amount': ['sum', 'mean'], 'transaction_id': 'count'})`. Hierarchical column names ko flatten kiya aur top-performing telecom circles aur plans ka ranking table prepare kiya.	**6 hrs** Certified: **60 hrs**
24 Oct 2026 Sat (Cal Day 13)	WORKING DAY 11 **Relational Data Merging, Joins & Concatenation**	**THEORY & CONCEPT (2.0H):** Relational database joins ko Pandas me apply karne ke methods padhe. Multi-table enterprise architecture (Customers, Plans, Transactions) me Joins ki zaroorat samjhi. 4 join types: **Inner Join**, **Left Outer Join**, **Right Outer Join**, aur **Full Outer Join** ke mathematical Venn-diagrams study kiye. `pd.concat()` ke axis=0 aur axis=1 mechanics samjhe. **HANDS-ON LAB & CODE (4.0H):** 3 alag-alag datasets load kiye: Customers (50k rows), Transactions, aur Plans Master. `pd.merge(transactions, customers, on='customer_id', how='left')` se customer demographics merge kiye; fir plan specifications ko `plan_id` par join kiya. Suffix overlaps (_x, _y) clean karke Master Analysis Table banaya.	**6 hrs** Certified: **66 hrs**
25 Oct 2026 Sun (Cal Day 14)	SUNDAY HOLIDAY	***Weekly Academic Off:*** *Self-study, revision of previous week concepts & SPU logbook maintenance.*	**0 hrs** Certified: 66 hrs

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26 Oct 2026 Mon (Cal Day 15)	WORKING DAY 12 **Pivot Tables & Wide-to-Long Reshaping (Melt)**	**THEORY & CONCEPT (2.0H):** Multi-dimensional business reporting ke liye **Pivot Tables** aur reshaping operations explore kiye. `pd.pivot_table()` me multi-index rows, columns, aur aggregation margins (Grand Totals) design karna samjha. Wide format data (months as columns) ko unpivot karke analysis-friendly long format me convert karne ke liye `pd.melt()` ka algorithmic mechanism study kiya. **HANDS-ON LAB & CODE (4.0H):** `pd.pivot_table(df, values='amount', index='operator', columns='payment_mode', aggfunc='sum', fill_value=0, margins=True)` run karke Operator vs Payment Mode ka Cross-Tabular Matrix banaya. Wide financial summary ko `pd.melt(id_vars=['operator'], value_vars=['Oct_Revenue', 'Nov_Revenue'])` se long format me transform kiya.	**6 hrs** Certified: **72 hrs**
27 Oct 2026 Tue (Cal Day 16)	WORKING DAY 13 **DateTime Parsing & Temporal Time-Series Analysis**	**THEORY & CONCEPT (2.0H):** Temporal data analytics aur time-series handling ke concepts padhe. Timestamps ko string ki jagah `datetime64[ns]` me parse karne ki importance samjhi taaki date arithmetic aur lag calculations possible hon. Date components extraction (`.dt.day_name()`, `.dt.hour`) aur business periodicity ke according frequency resampling (`resample('D')`, `resample('W')`) study kiye. **HANDS-ON LAB & CODE (4.0H):** `pd.to_datetime(df['recharge_time'])` execute karke transaction timestamps parse kiye. Feature engineering columns banaye: `Hour_of_Day`, `Day_of_Week`, aur `Is_Weekend`. Recharge date index par set karke daily revenue velocity calculate ki. Paya gaya ki month-ends aur weekends par recharge activity me 35% spike aata hai.	**6 hrs** Certified: **78 hrs**
28 Oct 2026 Wed (Cal Day 17)	WORKING DAY 14 **Capstone Project: Telecom Dataset & Synopsis Design**	**THEORY & CONCEPT (2.0H):** College capstone project ke formal scope, problem statement, aur business objectives ko finalize kiya. Project title select kiya: **'TELEPULSE 360°: Pan-India Telecom Intelligence, ARPU Optimization & Consumer Churn Platform'**. 5 core research questions formulate kiye (Recharge cycle decay, payment gateway drop-offs, ARPU across Jio/Airtel/Vi/BSNL, 4G/5G booster trends). **HANDS-ON LAB & CODE (4.0H):** 50,000+ transaction records ka relational schema validate kiya. Har variable ki definition, data type, allowed range, aur foreign key constraints document karke formal **Data Dictionary** banayi. College presentation aur faculty evaluation ke liye 2-page ka official **Project Synopsis** draft kiya.	**6 hrs** Certified: **84 hrs**
29 Oct 2026 Thu (Cal Day 18)	WORKING DAY 15 **Outlier Analysis (IQR / Z-Score) & Cleaning Pipeline**	**THEORY & CONCEPT (2.0H):** Statistical outlier detection techniques study ki jo data distribution ko clean karne me use hoti hain. **Z-Score** method (standard normal distribution, threshold $\pm 3\sigma$) aur **IQR (Interquartile Range) Method** ($Q_1 - 1.5 \times IQR$ to $Q_3 + 1.5 \times IQR$, skewed business data ke liye robust) ka mathematical comparison kiya. **HANDS-ON LAB & CODE (4.0H):** Numeric columns ('amount', 'delay_days', 'data_consumed') par IQR boundaries calculate kiye. Identified extreme values ko verify kiya ki wo genuine high-value annual recharges (₹2999) hain ya system errors. End me ek modular Python function `clean_dataset_pipeline(raw_df)` likha jo raw data input lekar fully cleaned output return karta hai.	**6 hrs** Certified: **90 hrs**
30 Oct 2026 Fri (Cal Day 19)	WORKING DAY 16 **Data Visualization Architecture with Matplotlib**	**THEORY & CONCEPT (2.0H):** Python foundational plotting library **Matplotlib** aur uske **Object-Oriented Architecture** ko study kiya. `pyplot` state-machine interface vs `Figure` & `Axes` OOP approach ke beech ka farak samjhaya. Visual hierarchy, color theory, labels, tick locators, legends, aspect ratio, aur chart integrity ke principles samjhe. **HANDS-ON LAB & CODE (4.0H):** `fig, ax = plt.subplots(figsize=(10,6))` structure use karke Line Plots (Monthly revenue trend), Bar Charts (Operator sales), aur Histograms (Recharge amount distribution) banaye. Customizations implement kiye: `ax.grid(True)`, custom palettes, aur bar-top annotations. Charts ko publication-quality resolution (`dpi=300`) me export kiya.	**6 hrs** Certified: **96 hrs**
31 Oct 2026 Sat (Cal Day 20)	COLLEGE INTERNAL EXAMINATION	***Academic / Public Off:*** *No company training sessions scheduled (College Internal Examination).*	**0 hrs** Certified: 96 hrs
01 Nov 2026 Sun (Cal Day 21)	SUNDAY HOLIDAY	***Weekly Academic Off:*** *Self-study, revision of previous week concepts & SPU logbook maintenance.*	**0 hrs** Certified: 96 hrs
02 Nov 2026 Mon (Cal Day 22)	WORKING DAY 17 **Statistical Visualizations with Seaborn**	**THEORY & CONCEPT (2.0H):** Matplotlib ke upar bani high-level statistical visualization library **Seaborn** ko study kiya. Seaborn direct Pandas DataFrames ke sath integrate hoti hai aur automatically statistical aggregations, confidence intervals, aur semantic mappings (hue, `size`) handle karti hai. Distribution plots aur Categorical comparison plots ke statistical significance ko samjha. **HANDS-ON LAB & CODE (4.0H):** `sns.set_theme(style='whitegrid')` apply kiya. Alag-alag operators ke recharge ticket size compare karne ke liye Boxplots banaye (`sns.boxplot(x='operator', y='amount', data=df)`) jisse median aur quartiles clear dikhe. Recharge frequency density ke liye KDE plot aur payment modes distribution ke liye `sns.countplot` visualize kiya.	**6 hrs** Certified: **102 hrs**

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03 Nov 2026 Tue (Cal Day 23)	WORKING DAY 18 **Multivariate Analysis, Covariance & Heatmaps**	**THEORY & CONCEPT (2.0H):** Multiple numerical variables ke aapsi correlation ko quantify karne ke liye **Pearson Correlation Coefficient (r)** ka concept samjha (range -1 to +1). Correlation vs Causation ka difference study kiya. Multicollinearity kya hoti hai aur correlation matrix ke visual representation ke liye annotated Heatmaps aur Facet Grids ka use kaise hota hai, ispar in-depth study ki. **HANDS-ON LAB & CODE (4.0H):** Numerical features ka correlation matrix calculate kiya (`corr_matrix = df[['amount', 'validity', 'daily_data', 'recharge_gap']].corr()`). Seaborn se stunning diverging heatmap plot kiya (`sns.heatmap(corr_matrix, annot=True, cmap='coolwarm')`). Paya gaya ki validity aur amount me high correlation ($r=0.88$) hai. Insights document kiye.	**6 hrs** Certified: **108 hrs**
04 Nov 2026 Wed (Cal Day 24)	WORKING DAY 19 **Comprehensive Exploratory Data Analysis (EDA) Notebook**	**THEORY & CONCEPT (2.0H):** Pure EDA methodology ko end-to-end structure kiya: **Univariate Analysis** (single variable characteristics), **Bivariate Analysis** (pairwise relationships), aur **Multivariate Analysis** (multi-dimensional interactions). Har chart ke neeche formal business interpretation aur executive observations likhne ke guidelines samjhe. **HANDS-ON LAB & CODE (4.0H):** Professional Jupyter Notebook (`01_telecom_eda.ipynb`) me 50,000 records par comprehensive EDA run kiya. Top findings: (1) 68% recharges UPI se huye, (2) 28-day plan users 3-4 din ka gap lete hain renewal me, (3) Jio me micro data boosters volume highest hai, (4) Airtel ka ARPU highest (₹214) hai. Markdown commentary compile ki.	**6 hrs** Certified: **114 hrs**
05 Nov 2026 Thu (Cal Day 25)	WORKING DAY 20 **SQL Foundations for Data Analysts & SQLite Modeling**	**THEORY & CONCEPT (2.0H):** Data Analysis ke core skill—**Structured Query Language (SQL)**—ki shuruat ki. Relational Database Management Systems (RDBMS), ACID properties, Table Schemas, Primary Keys, Foreign Keys, aur Entity-Relationship (ER) modeling study ki. Enterprise production me CSVs ki jagah relational SQL databases kyu use hote hain, iski theoretical importance samjhi. **HANDS-ON LAB & CODE (4.0H):** Python ke `sqlite3` aur `SQLAlchemy` ka use karke local database `telecom_warehouse.db` create kiya. Cleaned DataFrames ko relational tables (`customers`, `plans`, `recharge_transactions`) me dump kiya. Basic DDL/DQL queries run ki: `SELECT * FROM recharge_transactions LIMIT 10` aur `SELECT DISTINCT operator FROM customers`.	**6 hrs** Certified: **120 hrs**
06 Nov 2026 Fri (Cal Day 26)	WORKING DAY 21 **SQL Filtering, Sorting & Pagination (Dhanteras Working Day)**	**THEORY & CONCEPT (2.0H):** SQL me conditional query filtering aur result ordering ke mechanics deeply practice kiye. `WHERE` clause me logical operators (`AND`, `OR`, `NOT`), range evaluation (`BETWEEN`), set membership (`IN`), aur text wildcards (`LIKE '%4G%'`) ke rules samjhe. Pagination ke liye `ORDER BY`, `LIMIT`, aur `OFFSET` clauses ka use padha. **HANDS-ON LAB & CODE (4.0H):** Targeted business queries execute ki: UPI payment mode se kiye gaye ₹500 se bade successful transactions retrieve kiye; Jio aur Airtel ke 84-day plans filter kiye; transactions ko amount ke descending order me sort karke top 20 highest-paying consumers extract kiye (`ORDER BY amount DESC LIMIT 20`). Results ko Pandas DataFrame me fetch kiya.	**6 hrs** Certified: **126 hrs**
07 Nov 2026 Sat (Cal Day 27)	DIWALI HOLIDAY (DAY 1 OF 4: NARAK CHATURDASHI)	***Academic / Public Off:*** *No company training sessions scheduled (Diwali Holiday (Day 1 of 4: Narak Chaturdashi)).*	**0 hrs** Certified: 126 hrs
08 Nov 2026 Sun (Cal Day 28)	SUNDAY HOLIDAY (DIWALI / LAXMI PUJAN - DAY 2 OF 4)	***Weekly Academic Off:*** *Self-study, revision of previous week concepts & SPU logbook maintenance (Diwali / Laxmi Pujan - Day 2 of 4).*	**0 hrs** Certified: 126 hrs
09 Nov 2026 Mon (Cal Day 29)	DIWALI HOLIDAY (DAY 3 OF 4: GOVARDHAN PUJA)	***Academic / Public Off:*** *No company training sessions scheduled (Diwali Holiday (Day 3 of 4: Govardhan Puja)).*	**0 hrs** Certified: 126 hrs
10 Nov 2026 Tue (Cal Day 30)	DIWALI HOLIDAY (DAY 4 OF 4: BHAI DOOJ)	***Academic / Public Off:*** *No company training sessions scheduled (Diwali Holiday (Day 4 of 4: Bhai Dooj)).*	**0 hrs** Certified: 126 hrs
11 Nov 2026 Wed (Cal Day 31)	WORKING DAY 22 **SQL Aggregations & GROUP BY / HAVING (Post-Diwali Session)**	**THEORY & CONCEPT (2.0H):** Post-Diwali session me SQL aggregate functions aur grouping mechanics deeply study kiye. `COUNT()`, `SUM()`, `AVG()`, `MIN()`, `MAX()` ke handling rules dekhe. **`WHERE` vs `HAVING` clause ka difference**: `WHERE` individual rows par filter lagata hai grouping se pehle, jabki `HAVING` aggregated summary values par filter lagata hai grouping ke baad. **HANDS-ON LAB & CODE (4.0H):** Complex enterprise analytical query likhi: `SELECT circle, operator, COUNT(transaction_id) AS total_recharges, ROUND(AVG(amount), 2) AS arpu, SUM(amount) AS total_revenue FROM recharge_transactions WHERE status = 'SUCCESS' GROUP BY circle, operator HAVING total_revenue > 100000 ORDER BY total_revenue DESC;`. Circle-wise revenue distribution analyze kiya.	**6 hrs** Certified: **132 hrs**

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12 Nov 2026 Thu (Cal Day 32)	WORKING DAY 23 **Dashboard Making Skill: Design Systems & Multi-Table Relational Project**	**DASHBOARD MAKING SKILL (2.0H):** Modern Business Intelligence & Web Dashboard Design Systems study kiye. Information architecture, F-pattern visual reading flow, 60-30-10 color rule for financial metrics, and typography hierarchy design kiya. Streamlit UI layout principles (`st.set_page_config(layout='wide')`, responsive containers) ka foundation establish kiya. **CAPSTONE PROJECT (4.0H):** 50,000+ Telecom records ka 3-Table Relational Join execute kiya: `SELECT c.customer_id, c.operator, c.circle, p.plan_name, t.amount, t.payment_mode FROM recharge_transactions t INNER JOIN customers c ON t.customer_id = c.customer_id INNER JOIN plans p ON t.plan_id = p.plan_id WHERE t.status = 'SUCCESS';`. Multi-table foreign keys validate kiye.	**6 hrs** Certified: **138 hrs**
13 Nov 2026 Fri (Cal Day 33)	WORKING DAY 24 **Dashboard Making Skill: Reactive Filtering & Advanced SQL Window Functions**	**DASHBOARD MAKING SKILL (2.0H):** Dashboard me Reactive State Management aur Dynamic Sidebar Controls create karna seekha. `st.sidebar.selectbox` (All Operators vs Jio vs Airtel vs Vi vs BSNL), dynamic multi-select circle filters, aur date-range pickers ka live binding code likha jo bina page refresh ke instant reactive data slices return karta hai. **CAPSTONE PROJECT (4.0H):** Customer churn aur recharge delay gap calculate karne ke liye `LAG()` window function use kiya: `LAG(recharge_time) OVER (PARTITION BY customer_id ORDER BY recharge_time)`. Plan expiry ke baad renewal gap > 7 days wale 3,840 high-risk churn subscribers identify kiye.	**6 hrs** Certified: **144 hrs**
14 Nov 2026 Sat (Cal Day 34)	WORKING DAY 25 **Dashboard Making Skill: Caching & High-Performance Data Pipeline**	**DASHBOARD MAKING SKILL (2.0H):** Big Data Web Dashboards me latency optimization seekha. Streamlit ka memory caching decorator `@st.cache_data(ttl=3600)` implement kiya taaki 50,000+ rows baar-baar disk se read na hon. CSV vs Parquet format benchmark kiya (Parquet ingestion 6.8x faster aur memory footprint 55% lower paya gaya). **CAPSTONE PROJECT (4.0H):** Automated Python ETL Pipeline (`etl_pipeline.py`) develop kiya jo raw telecom transaction logs ko parse, clean, type-cast karke `telecom_warehouse.db` SQLite warehouse me auto-load karta hai. Data quality validation test cases pass kiye.	**6 hrs** Certified: **150 hrs**
15 Nov 2026 Sun (Cal Day 35)	SUNDAY HOLIDAY	***Weekly Academic Off:*** *Self-study, revision of previous week concepts & SPU logbook maintenance.*	**0 hrs** Certified: 150 hrs
16 Nov 2026 Mon (Cal Day 36)	WORKING DAY 26 **Dashboard Making Skill: Executive KPI Metric Cards & ARPU Engine**	**DASHBOARD MAKING SKILL (2.0H):** Executive C-Suite KPI Metric Cards design karne ka skill seekha: `st.metric(label, value, delta)`. Custom CSS styling apply karke glassmorphism KPI tiles banaye jisme Gross Merchandise Value (GMV), Industry ARPU, Total Active Subscribers, aur Transaction Success Rate % positive/negative trend arrows ke sath render huye. **CAPSTONE PROJECT (4.0H):** Operator-wise competitive benchmarking SQL analytics code compile kiya: Airtel (ARPU: ₹214), Jio (ARPU: ₹188, Volume Leader: 44%), Vi (ARPU: ₹142, Churn Risk: 18%), aur BSNL (ARPU: ₹104). MoM revenue growth rate matrices export kiye.	**6 hrs** Certified: **156 hrs**
17 Nov 2026 Tue (Cal Day 37)	WORKING DAY 27 **Dashboard Making Skill: Interactive Plotly Visualizations & Funnel Diagnostics**	**DASHBOARD MAKING SKILL (2.0H):** Advanced interactive data visualization engineering with Plotly Express & Graph Objects. Hover data formatting (custom tooltip templates), dual-axis plotting (Revenue Bars + Margin Line), dynamic chart zooming, aur color gradient scaling ke production patterns master kiye. **CAPSTONE PROJECT (4.0H):** Fintech Payment Gateway reliability diagnostics complete kiye: UPI vs Debit Cards vs NetBanking vs Wallets ka Interactive Drop-off Funnel Chart (Initiated -> Gateway Handoff -> Bank Auth -> Success) banaya. Peak-hour server timeout spike (8:30 PM to 10:30 PM) quantify kiya.	**6 hrs** Certified: **162 hrs**
18 Nov 2026 Wed (Cal Day 38)	WORKING DAY 28 **Dashboard Making Skill: Geospatial Mapping & RFM-D ML Clustering**	**DASHBOARD MAKING SKILL (2.0H):** Geospatial Telecom Circle Mapping skill seekha: Plotly Choropleth Map of India integrate kiya. State/Circle boundary GeoJSON data bind karke color intensity ke basis par recharge density aur operator market share show kiya. Hover karne par state-specific telecom metrics render huye. **CAPSTONE PROJECT (4.0H):** Telecom RFM-D (Recency, Frequency, Monetary + Daily Data Intensity) segmentation model code kiya. Scikit-learn se K-Means Clustering ($k=4$) train kiya with `StandardScaler`, Elbow Curve, aur Silhouette Score evaluation. 3D Plotly cluster scatter view prepare kiya.	**6 hrs** Certified: **168 hrs**
19 Nov 2026 Thu (Cal Day 39)	WORKING DAY 29 **Dashboard Making Skill: Multi-Page App Assembly & Navigation**	**DASHBOARD MAKING SKILL (2.0H):** Production Streamlit Multi-Page Directory Architecture build ki: `pages/1_` 📈 `_Executive_Overview.py`, `pages/2_` 🗺️ `_Circle_Geospatial.py`, `pages/3_` 👥 `_Customer_Segmentation.py`, aur `pages/4_` 🎯 `_Smart_Simulator.py`. Clean navigation bar aur consistent branding apply kiye. **CAPSTONE PROJECT (4.0H):** Main application `app.py` assemble ki. Sabhi analytical pipelines, SQL queries, ML models aur Plotly charts ko real-time reactive filters ke sath connect karke end-to-end local testing perform ki. Responsive testing verify ki.	**6 hrs** Certified: **174 hrs**

DATE & DAY (2026)	TRAINING STATUS / TOPIC	DAILY WORK REPORT LOG (THEORY + HANDS-ON LAB / DASHBOARD SKILL + PROJECT)	HOURS LOGGED
20 Nov 2026 Fri (Cal Day 40)	WORKING DAY 30 (180H TARGET MET!) **Dashboard Making Skill: What-If Simulators & 180h Certified Milestone**	**DASHBOARD MAKING SKILL (2.0H):** Interactive What-If Decision Simulators design karne ka skill seekha: `st.slider` aur mathematical sensitivity models connect karke dynamic prediction widgets banaye. User input change hote hi real-time financial ROI recalculate karne ka reactive engine banaya. **CAPSTONE PROJECT (4.0H - 180H TARGET MET!):** Do interactive decision engines deploy kiye: (1) **Smart Plan Recommender** jo consumer usage ke hisaab se optimal cost-saving plan recommend karta hai; (2) **Tariff Hike Simulator** jo 10%-25% tariff increase par churn loss vs net profit calculate karta hai. **Mandatory 180.0 Certified OJT Hours achieve huye!**	**6 hrs** Certified: **180 hrs**
21 Nov 2026 Sat (Cal Day 41)	WORKING DAY 31 **Dashboard Making Skill: Streamlit Cloud Deployment & GitHub Portfolio**	**DASHBOARD MAKING SKILL (2.0H):** Cloud Web Application Deployment & DevOps workflow seekha. `requirements.txt` version lock, `.streamlit/config.toml` production theme settings, aur **Streamlit Community Cloud** par 1-click free cloud hosting execute karke public live portfolio URL generate kiya. **CAPSTONE PROJECT (4.0H):** Public GitHub repository structure build kiya: modular code (`src/`), Jupyter notebooks, sample data, architecture diagrams, live web app badges, aur comprehensive `README.md`. SPU Presentation Deck (12-slide PPT) finalize kiya.	**6 hrs** Certified: **180 hrs**
22 Nov 2026 Sun (Cal Day 42)	SUNDAY HOLIDAY	***Weekly Academic Off:*** *Self-study, revision of previous week concepts & SPU logbook maintenance.*	**0 hrs** Certified: 180 hrs
23 Nov 2026 Mon (Cal Day 43)	WORKING DAY 32 **SPU Official Report Dossier (10 Standard Sections) & Faculty Sign-Off**	**DASHBOARD MAKING SKILL (2.0H):** Executive Technical Presentation & Viva Voce Defense skills: Dashboard walkthrough rehearsal, handling technical questions (Streamlit caching, SQL window functions, K-Means mathematical convergence, OLTP vs OLAP architecture), aur faculty evaluation rubrics practice ki. **CAPSTONE PROJECT (4.0H):** SPU circular format ke according formal **35+ Page OJT Project Report** complete kiya (10 Standard Sections: Coverpage, Certificates, Acknowledgement, Index, Introduction, Objectives, Learning Experience, Outcomes, Conclusions, Attachments). Company Mentor & Faculty sign-off ready kiya.	**6 hrs** Certified: **180 hrs**
24 Nov 2026 Tue (Cal Day 44)	GURU NANAK JAYANTI (OFFICIAL SPU SEM 7 DEADLINE & COMPLETION DAY)	***Academic / Public Off:*** *No company training sessions scheduled (Guru Nanak Jayanti (Official SPU Sem 7 Deadline & Completion Day)).*	**0 hrs** Certified: 180 hrs

SPU Official Report Format (10 Standard Sections)

1. **Coverpage:** Student Name, Enrollment No, SPU Sem 7, Company Name & Logos
2. **Certificate:** Company Completion Certificate with Seal & Guide Sign
3. **Acknowledgement:** Faculty Guide, HOD, and Company Mentor
4. **Table Of Contents:** Structured chapter index with page numbers
5. **Introduction:** Data Analysis domain, Industry context & Telecom sector
6. **Objective Of The OJT:** 180h practical competency, ETL, and Dashboarding
7. **Learning Experience:** Day 1-22 Python/SQL mastery + Last 10 Days Dashboard skills
8. **Learning Outcomes:** TELEPULSE 360° Platform, 12 Dynamic Features, GitHub CI/CD
9. **Conclusions / Summary:** Business impact, ARPU findings, and future predictive scope
10. **Attachment:** Code listings, ER diagrams, Dashboard screenshots, Viva slides

Capstone Project: TELEPULSE 360° (50,000+ Transactions)

Project Title: TELEPULSE 360°: Pan-India Telecom Intelligence, ARPU Optimization, Payment Gateway Reliability & Consumer Churn Platform
Tech Stack: Python 3.x, SQLite/SQLAlchemy, Pandas, Scikit-Learn, Plotly Express, Streamlit Cloud
Multi-Operator Scale: 50,000+ Records across Reliance Jio, Bharti Airtel, Vodafone Idea (Vi), and BSNL across 22 Indian Telecom Circles
Key Dynamic Features: ARPU Benchmarking, 28 vs 84-day Cycle Decay, Micro-Booster Spikes, Payment Gateway Drop-Off Funnel, Telecom RFM-D 3D Clustering, Live K-Means Tuner, Pan-India Choropleth Map, What-If Tariff Hike Simulator
Evaluation Deliverables: SPU 10-Section Report Dossier + 12-Slide Defense PPT + Live Streamlit Cloud URL + Daily Signed Logbook

Student Signature	Company Mentor / Supervisor	SPU Faculty Coordinator / HOD
Intern Trainee (Candidate)	Authorized Seal & Signature	Department Evaluation Committee